b) Conjugate Gradient method.

1. 2 epochs are needed
   Because there are 2 weights.

2.

Epoch 1 is identical to 2.a. $\Rightarrow \nabla E(w(0)) = [-1, -4]$

$h_0 = -\nabla E = [1, 4]$

Epoch 2 $\Rightarrow$ train begin ~ in our :

$$\frac{\partial E}{\partial w_1} = 2(0.7546) - 2 = -0.4908$$

$$\frac{\partial E}{\partial w_2} = 4(2.0203) - 8 = 0.1212$$

$$\nabla E = [-0.4908, 0.1212] \Rightarrow -\nabla E = [0.4908, -0.1212]$$

Now Conjugate

$\hookrightarrow h_1 = -\nabla E(w(1)) + \gamma \cdot h_0$

$\hookrightarrow \gamma = \dfrac{|\nabla E(w(1))|^2}{|\nabla E(w(0))|^2}$

Fletcher - Reeves !

$$= \frac{0.2497}{17} = 0.0147$$

Therefore $h_1 = [0.4945, -0.0824]$